

CO5-AT1
MLA0101-FUNDAMENTALS OF MACHINE LEARNING

Annexure A
Constraint Based learning

Question 1 (20 Marks): Reinforcement Learning for Warehouse Robot Navigation Scenario

An e-commerce company uses an autonomous warehouse robot to transport packages between storage racks and dispatch areas. The robot must reach the destination while avoiding obstacles, minimizing travel time, and conserving battery power.

Constraints

- The robot can move only in four directions (Up, Down, Left, Right).
- Obstacles cannot be crossed.
- Each movement consumes one unit of battery.
- The robot must reach the destination within 25 moves.
- Battery capacity is limited to 30 units.
- Some paths provide positive rewards, while blocked paths incur penalties.
- The environment is stochastic, and some actions may not produce the intended movement.

Task

Design a Reinforcement Learning solution by selecting an appropriate exploration strategy and applying either Value Iteration or Policy Iteration to determine the optimal navigation policy. Justify your approach based on the given constraints.

Question 2 (20 Marks): Multi-Armed Bandit for Online Advertisement Selection Scenario

A digital marketing company displays one of several online advertisements to website visitors. The objective is to maximize the click-through rate while learning customer preferences over time.

Constraints

- Five advertisements are available.
- Only one advertisement can be displayed per visitor.
- The click probability of each advertisement is initially unknown.
- The system must balance exploration and exploitation.
- The total advertising budget allows only 10,000 user interactions.
- The solution should minimize cumulative regret while maximizing total rewards.

Task

Formulate the problem as a K-Armed Bandit problem. Select an appropriate exploration strategy (e.g., ϵ -greedy, UCB, or Thompson Sampling) and justify how it satisfies the given constraints. Explain how the reward mechanism influences advertisement selection.

Question 3 (20 Marks): Sampling Strategy for Machine Learning Model Development Scenario

A healthcare analytics company is developing a disease prediction model using a dataset containing one million patient records. Limited computational resources prevent training on the complete dataset.

Constraints

- Training memory is limited to 8 GB RAM.
- The sample must accurately represent all patient categories.
- The prediction accuracy should remain above 95%.
- Model confidence must be at least 95%.
- The solution should minimize sampling bias and computational cost.
- The selected learning approach should satisfy sample complexity and generalization requirements.

Task

Recommend an appropriate sampling method (e.g., Simple Random Sampling, Stratified Sampling, or Monte Carlo Sampling) and justify your choice. Analyze how sample complexity, VC dimension, Occam's Learning Principle, and accuracy-confidence boosting influence the model's performance under the given constraints.

1. REINFORCEMENT LEARNING FOR WAREHOUSE ROBOT NAVIGATION

1. Problem Understanding

The warehouse robot has to navigate from a start position to a destination while:

- Moving only Up, Down, Left, Right
- Avoiding obstacles
- Minimizing travel time
- Conserving battery
- Reaching the destination within 25 moves
- Using at most 30 battery units
- Handling positive rewards and penalties
- Working in a stochastic environment, where an action may not always result in the intended movement.

This can be formulated as a Markov Decision Process (MDP).

MDP Components

Component	Description
States (S)	Robot's position and remaining battery/moves
Actions (A)	Up, Down, Left, Right
Transition	Probability of reaching the intended next state
Reward (R)	Positive for useful movement/goal, negative for obstacles and battery usage
Policy (π)	Strategy deciding which action to take
Discount factor (γ)	Determines importance of future rewards

2. Reward Design

A suitable reward function is:

- **+100** → Reaching destination
- **+5** → Moving toward destination
- **-10** → Hitting/attempting an obstacle
- **-1** → Each movement
- **-20** → Wasting a movement or moving away from destination
- **-100** → Exceeding 25 moves or battery limit

The large positive reward for reaching the destination encourages the robot to complete its task. The movement penalty encourages the robot to find a short path.

3. Exploration Strategy

I would use ϵ -greedy exploration.

The robot chooses:

$$a = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \text{best known action} & \text{with probability } 1 - \epsilon \end{cases}$$

For example:

$$\epsilon = 0.1$$

means:

- 10% of the time → explore
- 90% of the time → exploit the best-known action.

A decaying ϵ is even better:

$$\epsilon_t = \max(\epsilon_{min}, \epsilon_0 \times decay^t)$$

Initially, the robot explores more. As it learns, it increasingly follows the best policy.

Why ϵ -greedy?

Because the warehouse environment is initially unknown and stochastic. The robot must explore alternative routes, but excessive exploration could waste battery and movements.

Therefore, decaying ϵ -greedy provides a good balance between exploration and exploitation.

4. Value Iteration

I would use Value Iteration to determine the optimal navigation policy.

The Bellman optimality equation is:

$$V(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s', | sa) V(s') \right]$$

Where:

- $V(s)$ = value of state s
- $R(s, a)$ = reward for taking action a
- $P(s', | sa)$ = probability of reaching next state s'
- γ = discount factor

Algorithm

1. Initialize $V(s) = 0$ for all states.
2. For every state, calculate the value of each action.
3. Select the action with maximum expected value.
4. Repeat until values converge.
5. Extract the optimal policy.

The policy is:

$$\pi^*(s) = \arg \max_a \left[R(s, a) + \gamma \sum_{s'} P(s', | sa) V(s') \right]$$

5. Handling Stochastic Actions

Suppose the robot selects Right.

There could be:

- 80% probability $\rightarrow$ move Right
- 10% probability $\rightarrow$ remain in the same position
- 10% probability $\rightarrow$ move in another valid direction

Therefore, Value Iteration considers the expected value of all possible outcomes, rather than assuming the action always succeeds.

This makes it suitable for the stochastic warehouse.

6. Constraint Handling

The robot's state can include:

$$s = (x, y, b, m)$$

where:

- x, y = position
- b = remaining battery
- m = number of moves used

A state becomes invalid when:

$$b < 0$$

or

$$m > 25$$

The optimal policy must therefore satisfy:

$$m \leq 25$$

and

$$b \leq 30$$

Since each movement consumes one battery unit:

$$\text{Battery Used} = \text{Number of Moves}$$

Thus, reaching the destination within 25 moves automatically satisfies the 30-unit battery constraint, assuming one battery unit per movement.

7. Final Answer

Selected approach:

Decaying ϵ -greedy exploration + Value Iteration

This is appropriate because:

- ϵ -greedy provides exploration.
- Decaying ϵ gradually shifts toward exploitation.
- Value Iteration handles stochastic transitions.
- Movement penalties encourage shorter routes.
- Obstacle penalties prevent unsafe routes.
- Goal rewards encourage reaching the destination.
- The state can include position, battery, and number of moves.
- The 25-move constraint ensures battery consumption remains within the 30-unit capacity.

Conclusion

The resulting policy directs the robot toward the destination while avoiding obstacles, minimizing unnecessary movement, conserving battery, and handling uncertainty in movement.

2 .MULTI-ARMED BANDIT FOR ONLINE ADVERTISEMENT SELECTION

1. Problem Formulation

This problem can be formulated as a 5-Armed Bandit problem.

There are five advertisements:

$$A = \{A_1, A_2, A_3, A_4, A_5\}$$

For every website visitor, the system selects exactly one advertisement.

The click probability of each advertisement is unknown.

For example:

Advertisement	True CTR
Ad 1	Unknown
Ad 2	Unknown
Ad 3	Unknown
Ad 4	Unknown
Ad 5	Unknown

The system learns the CTR from observed clicks.

2. Reward Mechanism

Define:

$$\text{Reward} = \begin{cases} 1 & \text{if user clicks the advertisement} \\ 0 & \text{if user does not click} \end{cases}$$

Therefore, if an advertisement receives:

- 100 displays
- 20 clicks

then its estimated CTR is:

$$\hat{p} = \frac{20}{100} = 0.20$$

or:
20%

The objective is:

$$\max \sum 10000 R_t$$

while minimizing cumulative regret.

3. Exploration Strategy

I would select **Upper Confidence Bound (UCB)**.

The UCB value for advertisement i is:

$$UCB_i = \hat{\mu}_i + \sqrt{\frac{2 \ln t}{n_i}}$$

Where:

- $\hat{\mu}_i$ = estimated CTR
- n_i = number of times advertisement i was displayed
- t = total number of interactions

The system selects:

$$A_t = \arg \max_i UCB_i$$

4. How UCB Balances Exploration and Exploitation

UCB contains two terms:

$$\underbrace{\hat{\mu}_i}_{\text{Exploitation}} + \underbrace{\sqrt{\frac{2 \ln t}{n_i}}}_{\text{Exploration}}$$

Exploitation

If an advertisement has a high estimated CTR, its first term becomes large.

Therefore, it is selected more frequently.

Exploration

If an advertisement has been selected only a few times, n_i is small.

Therefore:

$$\sqrt{\frac{2 \ln t}{n_i}}$$

becomes large.

This forces the algorithm to test advertisements that have not been sufficiently explored.

5. Example

Suppose at some point:

Ad	Estimated CTR	Number shown
A1	0.10	200
A2	0.15	100
A3	0.20	20
A4	0.08	500
A5	0.12	10

A5 has a lower estimated CTR than A3, but because it has been displayed only 10 times, its exploration bonus is large.

Therefore, UCB may select A5.

This prevents the algorithm from incorrectly rejecting a potentially high-performing advertisement because of insufficient observations.

6. Regret

If the best advertisement has true click probability: $\mu^* = 0.30$
and the algorithm selects an advertisement with:

$$\mu_i = 0.20$$

then the expected regret for that interaction is:

$$0.30 - 0.20 = 0.10$$

Cumulative regret over 10,000 interactions is:

$$R_T = \sum_{t=1}^{10000} (\mu^* - \mu_{A_t})$$

The objective is to minimize this value.

7. Why UCB Is Appropriate

UCB is suitable because:

- There are only 5 advertisements.
- Click probabilities are initially unknown.
- It automatically balances exploration and exploitation.
- It does not require manually choosing ϵ .
- Poorly explored advertisements receive additional opportunities.
- High-performing advertisements are increasingly exploited.
- It is computationally inexpensive for 10,000 interactions.
- It helps minimize cumulative regret.

Final Answer

Selected method: UCB

The system initially gives every advertisement opportunities to collect data. UCB then combines the estimated CTR with an uncertainty bonus.

As evidence accumulates, uncertainty decreases and the algorithm increasingly displays the advertisements with the highest expected click rate.

Thus, UCB maximizes total clicks while controlling exploration and cumulative regret over the **10,000-interaction budget**.

3.SAMPLING STRATEGY FOR MACHINE LEARNING MODEL DEVELOPMENT

1. Problem Understanding

The company has:

$$N = 1,000,000$$

patient records.

Training on the complete dataset is computationally expensive because of the 8 GB RAM limit.

The sample must:

- Represent all patient categories.
- Minimize sampling bias.
- Maintain accuracy above 95%.
- Provide at least 95% confidence.
- Reduce computational cost.
- Have adequate sample complexity.
- Generalize well to unseen data.

2. Selected Sampling Method

I recommend Stratified Random Sampling.

In stratified sampling, the population is divided into meaningful groups called **strata**, and samples are taken from each group.

For a disease prediction problem, possible strata could be:

- Disease positive
- Disease negative

Or additional patient categories such as:

- Age groups
- Gender categories
- Geographic groups
- Disease severity groups

The important point is that important categories are represented in the training sample.

3. Why Not Simple Random Sampling?

Simple Random Sampling selects records randomly from the entire population.

The problem is that rare categories may be underrepresented.

For example, suppose:

$$N = 1,000,000$$

and only:

$$20,000$$

patients belong to a rare disease category.

A random sample may contain too few rare cases.

This can cause:

- Sampling bias
- Poor minority-class performance
- Lower recall
- Poor generalization

Stratified sampling avoids this problem.

4. Sample Allocation

Suppose the population contains:

Category	Population
Disease Positive	100,000
Disease Negative	900,000
Total	1,000,000

If we choose a 10% stratified sample:

$$n = 100,000$$

then:

$$n_{positive} = 0.10(100,000) = 10,000$$

and:

$$n_{negative} = 0.10(900,000) = 90,000$$

Therefore:

$$n = 10,000 + 90,000 = 100,000$$

Both categories remain represented proportionally.

If a category is very rare, disproportionate stratified sampling can intentionally oversample it, followed by class weighting or appropriate evaluation.

5. Sample Complexity

Sample complexity describes how many training examples are required for a model to learn a good approximation of the target function.

A simplified learning-theory relationship is:

$$n = O\left(\frac{d + \ln(1/\delta)}{\epsilon^2}\right)$$

where:

- n = required sample size
- d = model complexity/VC dimension
- ϵ = desired generalization error
- δ = confidence failure probability

For **95% confidence**:

$$1 - \delta = 0.95$$

Therefore:

$$\delta = 0.05$$

and:

$$\ln(1/\delta) = \ln(20)$$

So increasing the sample size improves confidence and generalization.

6. VC Dimension

The **VC dimension** measures the capacity or complexity of a hypothesis class.

A model with a very high VC dimension can represent complicated decision boundaries, but generally requires more training data to generalize reliably.

Conceptually:

$$\text{Higher VC Dimension} \Rightarrow \text{Higher Sample Requirement}$$

Therefore, under an 8 GB RAM constraint, the company should avoid unnecessarily complex models. A model with suitable complexity can achieve good generalization using a smaller representative sample.

7. Occam's Learning Principle

Occam's Learning Principle states that, when multiple models explain the training data adequately, the simpler model is preferred.

Therefore: Do not use an unnecessarily complex model when a simpler model provides comparable predictive performance. A simpler hypothesis generally has lower effective complexity and can require fewer training examples. This is especially important because the company has limited computational resources.

8. Accuracy and Confidence

The company requires:

$$\text{Accuracy} > 95\%$$

and:

$$\text{Confidence} \geq 95\%$$

These are not exactly the same thing.

Accuracy

Accuracy measures how often predictions are correct:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

For example:

$$\frac{950}{1000} \times 100 = 95\%$$

Confidence

Confidence indicates how certain we are about an estimated performance or population parameter.

A 95% confidence interval can be written approximately as:

$$\hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Increasing n reduces the standard error:

$$SE \propto \frac{1}{\sqrt{n}}$$

Thus, larger representative samples generally provide more precise estimates.

9. Accuracy-Confidence Boosting

To improve reliability:

1. Use stratified sampling.
2. Ensure rare patient categories are represented.
3. Use an independent validation/test set.
4. Increase sample size if validation confidence intervals are too wide.
5. Use cross-validation where computationally feasible.
6. Tune the model on training data and evaluate only on held-out data.
7. Use an appropriately simple model to reduce overfitting.

The sample size should not be selected only because it fits into memory; it must also be large enough to satisfy the required generalization performance.

10. Why Stratified Sampling Is Best

Requirement	Stratified Sampling
1 million records	Reduces training dataset
8 GB RAM	Lower computational requirement
Represent all categories	✓ Excellent
Reduce sampling bias	✓
95% confidence	Can increase sample size as required
>95% accuracy	Depends on model/data; validate experimentally
Generalization	Better representation of population
Computational cost	Lower than using all records

Final Answer

Selected method: Stratified Random Sampling

Stratified sampling is the most appropriate because the dataset contains one million patient records and the sample must accurately represent all patient categories.

The population is divided into relevant strata, and random samples are selected from each stratum. This reduces the possibility that important or rare categories will be missing from the training set.

Sample complexity indicates that the required sample size depends on model complexity, desired accuracy, and confidence. Therefore, the company should choose a sample large enough to achieve the required generalization performance while remaining within the 8 GB RAM limitation.

VC dimension explains why unnecessarily complex models require more data. Occam's Learning Principle supports choosing a sufficiently simple model. Finally, validation, cross-validation, and confidence intervals can be used to verify whether the selected sample and model achieve the required **>95% accuracy and $\geq 95\%$ confidence**.

Therefore:

Stratified Random Sampling + appropriately controlled model complexity + independent validation provides the best balance between representativeness, computational cost, accuracy, and generalization.